

An Analytical Study of Demand Patterns and Price Ambiguity in a Construction Materials Retail Store

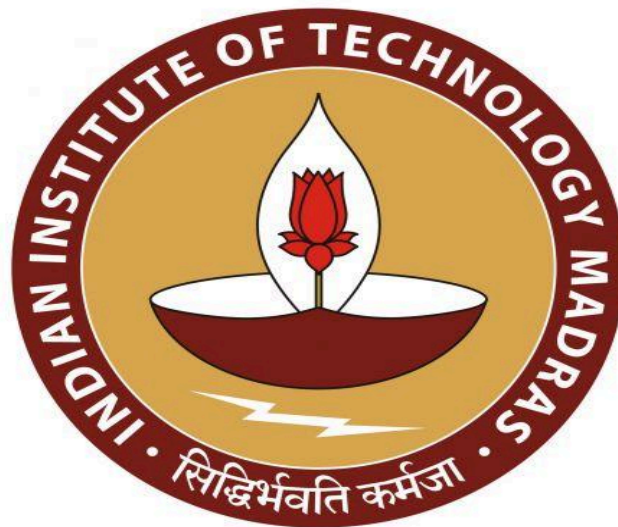
A Final Report for BDM Capstone Project

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Executive summary

Trishul Cement and Hardware Store is a small-scale construction materials retailer located on Jaganpura Road, Patna, Bihar, operating since 2000 under the proprietorship of Shreekanth Kumar. The store serves individual customers (~90% of sales) and small contractors, offering cements, TMT bars, stirrups, paints, primers, tarpaulin covers, and binding wire. Despite its long-standing presence, the business faces three critical operational challenges: (1) demand fluctuation driven by irregular bulk orders causing frequent stock imbalances, (2) inventory mismanagement resulting in stockouts or excess inventory, and (3) price ambiguity — an unstructured pricing strategy causing inconsistent profit margins that risk both customer retention and profitability.

The analysis was based on transaction-level data collected from sales records and owner-maintained logs. Descriptive statistics such as mean, standard deviation, and percentiles were used to understand demand patterns. Profit margin analysis, price vs. cost comparisons, and reorder point (ROP) calculations were applied to evaluate pricing efficiency and inventory performance. Visual tools such as scatter plots, box plots, and time-series graphs were used to support insights.

The findings reveal clear pricing ambiguity, particularly in cement and stirrup products, where the same product was sold at multiple prices within the same month, including instances of zero-margin sales. Frequent stockouts were observed in TMT bars (especially 10mm and 12mm), driven by reactive purchasing cycles where inventory repeatedly dropped below reorder levels before replenishment. Demand analysis shows high variability in cement, TMT bars, and stirrups, with sudden spikes making average-based planning unreliable, while paints and primers exhibit stable and predictable demand. These inefficiencies led to revenue leakage, inconsistent margins, and missed sales opportunities.

Based on these insights, recommendations include implementing minimum margin controls, standardizing pricing, introducing safety stock and reorder point systems,. These measures are expected to improve pricing decisions , reduce stockouts, and enhance overall profitability and operational efficiency.

Detailed Explanation of Analysis Method

1. Descriptive statistics

Descriptive statistics were used to understand how demand behaves for each product at Trishul Cement and Hardware Store. Since the store deals with construction materials measured in different units—cement in bags, TMT bars and binding wire in kilograms, stirrups and tarpaulin covers in pieces, and primer and paint in litres—direct comparison across categories can be misleading. Therefore, the analysis was conducted in two ways. First, a product-wise analysis was performed, where each product was studied within its own unit to understand its demand pattern independently. Second, for overall comparison across products, Z-score standardisation was applied to convert all values to a common scale, enabling meaningful inter-product comparison.

1.1 Descriptive Statistics (Product Wise)

Steps to Perform Descriptive Statistics (product wise) is as follows:

Step 1 : Separate Each Product's Data Using FILTER function

After data cleaning and preprocessing, the FILTER function in Google Sheets was applied to isolate quantity values for each of the 14 products from the cleaned dataset. All descriptive statistics were computed using the cleaned dataset, which includes zero-sales days, ensuring an accurate representation of daily demand patterns.

Step 2 : Computation of Product-wise Descriptive Statistics

1. Mean (Average Demand):The AVERAGE function was used to calculate the mean quantity sold per transaction.

Justification: Provides a baseline estimate of demand and a reference to assess variation.

2. Data Type Classification: Products were classified as discrete (bags, pieces) or continuous (kg, litres).

Justification: Ensures appropriate use of statistical measures and valid analysis.

3. Standard Deviation (Demand Fluctuation): The STDEV function measures deviation of demand from the mean.

Justification: Quantifies variability, directly addressing the demand fluctuation problem.

4. Minimum and Maximum Values: The MIN and MAX functions identified the lowest and highest demand levels.

Justification: Capture extreme demand conditions relevant for inventory planning.

5. Skewness (Distribution Shape): The SKEW function analyzes demand distribution; positive values indicate spikes.

Justification: Identifies irregular demand patterns and supports safety stock decisions.

6. Total Sales Days (Demand Frequency): The COUNTIF function was used to calculate the number of days on which each product recorded sales:

Justification: This measures how frequently a product is sold, helping distinguish between consistently demanded products and those with irregular or sporadic demand patterns, thereby supporting the analysis of demand fluctuation.

Step 3 : Product-wise Interpretation :

Each product's descriptive statistics were interpreted individually and compared only within the same product category. For example- within the cement category, UltraTech was compared with other cement brands (JSW, Nuvoco, Concreto) using descriptive statistics to examine differences in demand levels and variability. Similarly, within the TMT category, different sizes (8mm, 10mm, 12mm) were analyzed to understand variation in demand across sizes. In the case of stirrups, square, circular, and rectangular types were compared to observe differences in their demand patterns based on the computed statistics.

1.2 Z-score standardization

Z-score standardization was applied to enable comparison across products measured in different units (bags, kg, pieces, litres). Since direct comparison of raw demand values would be misleading, standardization transforms each product's demand into a common scale based on its mean and standard deviation. This allows evaluation of relative demand intensity and variability across products .

Formula Used :

$$Z = \frac{X - \mu}{\sigma}$$

Fig 1: Standardization Formula

Where ,

Z = Standard Score , X = Observed Value , μ = mean value , σ = standard deviation

Steps to Perform Z-score Standardization

Step 1 : Compute mean and standard deviation

Z-score standardisation was performed using the cleaned dataset, which includes zero-sales days to ensure accurate representation of demand.

For each product (e.g., Nuvoco cement), the mean and standard deviation were calculated using AVERAGE and STDEV function.

Step2 : Apply Z-score formula

For each observation, Z-score was calculated using the formula “=STANDARDIZE(cell, mean, std_dev) .”

Step3 : Repeat for all observations

The formula was applied across all rows to convert daily demand values into standardized values.

Worked Out Example:

		Cement(Bags)		
Date	Month	Nuvoco	Nuvoco Z	UltraTech
01-08-2025	Aug 2025	20	0.2498377677 ×	10
02-08-2025	Aug 2025	10	=STANDARDIZE(c4,6.15,15.41)	
03-08-2025	Aug 2025	0		0
04-08-2025	Aug 2025	30		15
05-08-2025	Aug 2025	0		0

Fig 2 : Z-score Calculation for Nuvoco Cement

From the cleaned dataset:

- Mean (μ) = 6.15
- Standard Deviation (σ) = 15.41
- Observed value (X) = 10 bags

Using the Z-score formula: $Z = (10 - 6.15) / 15.41 = 0.2498377677$

2. Reorder Point Analysis (ROP):

The Reorder Point (ROP) is the inventory level at which a new order should be placed to replenish stock before it runs out. It ensures that sufficient stock is available to meet demand during the supplier's lead time.

This method directly addresses the inventory management problem, where the store must decide when to reorder stock to avoid stockouts while also preventing overstocking.

Why ROP is Required for This Problem - The store experiences highly unpredictable demand patterns, where most days have low or zero sales but occasional bulk purchases create sharp

spikes. For eg. some days only 5 bags of JSW cement are sold, while on other days a single order picks up 100 bags at once This was observed from the descriptive analysis:

- High standard deviation → large variability in demand
- High positive skewness (>3) → presence of demand spikes
- Z-score analysis → extreme demand values on certain days

In such conditions, relying only on average demand would underestimate inventory requirements and lead to frequent stockouts during peak demand periods.

The ROP model solves this problem by combining:

- Expected demand during lead time
- A safety buffer to handle variability and spikes

This ensures that inventory levels are sufficient to meet both regular demand and unexpected fluctuations.

Mathematical Formula - $ROP = (\text{Mean Demand}(\mu) \times \text{Lead Time}(L)) + \text{Safety Stock}$

Explanation of the Formula Components

- **μ (Mean Demand):**

Average daily demand calculated from the cleaned dataset (including zero-sales days). It represents the expected demand level.

- **L (Lead Time):**

Number of days required for stock to arrive after placing an order . Based on store operations and supplier delivery patterns, products typically take **around 5 days** to reach the store, so **L = 5 days** was used in the analysis. During these 5 days, the store continues selling without any incoming stock. Therefore, the reorder point must cover expected demand during this period, along with a safety buffer. Given the high demand variability and spikes observed, including lead time ensures that stock does not run out while waiting for replenishment.

- **Safety Stock**

Since the demand data is non-normal and skewed, a traditional z-score based approach is not appropriate. Therefore, a data-driven empirical percentile approach was used, based on the cleaned dataset.

To capture demand variability and determine an appropriate service level, two percentiles were calculated:

1. 90th Percentile - The 90th percentile implies that the store can meet demand in approximately 90% of observed cases, meaning stockouts may occur only during extreme demand situations (around 10% of cases).
2. 95th percentile - The 95th percentile implies that the store can meet demand in 95% of demand scenarios, but at the cost of significantly higher inventory levels. .

Both percentiles were evaluated to understand the trade-off between demand coverage (service level) and inventory cost. While 95th percentile reduces the risk of stockouts further by covering a higher proportion of demand scenarios, it leads to excessive inventory. In contrast, 90th percentile provides a balanced solution by covering most demand situations while keeping inventory levels practical. Therefore, P90 was selected as the final service level for this business.

Steps for ROP Analysis:

Step 1 . Calculate Mean Demand (μ) from the cleaned dataset using AVERAGE function

Step 2 . Determine Lead Time (L)

Number of days required for replenishment (here, $L=5$)

Step 3 . Calculate Percentiles (90th percentile and 95th percentile) using cleaned data

Formula Used : =PERCENTILE(Range, Percentile)

Step 4 . Compute Safety Stock for both percentiles

Formula Used: Safety stock = 90th percentile - mean

Safety stock = 95th percentile - mean

Step 5 . Calculate Reorder Point

Worked Out Example: The following example illustrates the calculation of safety stock and reorder point for **Nuvoco Cement** using the empirical percentile approach based on the cleaned dataset (cell 187 to 197).

mean	6.14673913
P95	38.5
P90	15.7
safety stock 90	9.55326087
safety stock 95	32.35326087
avg usage	30.73369565
ROP 90 Percent	40.28695652
ROP 95 Percent	63.08695652

Fig 3: Safety Stock and ROP Calculation for Nuvoco Cement

Given , Mean demand (μ) = 6.1467 units/day , Lead time (L) = 5 days , P90 = 15.7 , P95 = 38.5

Step1: Calculation of Lead Time= $\mu \times L = 6.1467 \times 5 = 30.7337$

Step2: Safety Stock(90th percentile)= $15.7 - 6.1467 = 9.553$

Safety Stock(95th percentile)= $38.5 - 6.1467 = 32.3533$

Step3:

$$\text{ROP (90th percentile)} = 30.7337 + 9.5533 = 40.2869$$

$$\text{ROP (95th percentile)} = 30.7337 + 32.3533 = 63.0869$$

3. Profit Margin Analysis

Profit margin analysis is used to understand how much profit the store earns on each product by comparing its selling price with its cost price and expressing it as a percentage. In the case of **Trishul Cement and Hardware Store**, this analysis becomes important because of **price ambiguity**, where pricing decisions are not always consistent or clearly linked to costs.

Understanding Price ambiguity in “ Trishul Cement and Hardware Store-

Zero-margin sales:

Some products were sold at no profit ($SP = CP$), indicating lack of pricing control.

Declining margins over time:

Profit margins decreased even when cost remained constant, suggesting unplanned price reductions.

Delay in price updates:

Cost increased but selling price was not updated immediately, reducing margins.

Unexplained price changes:

Selling prices changed without clear cost justification.

Erratic margin variation:

Margins fluctuate significantly across similar transactions.

How Profit Analysis solves these problems

Profit margin analysis solves pricing issues by converting each transaction into a clear measure of profitability, making inconsistencies easy to detect. Zero-margin sales are

immediately identified as 0%, highlighting underpricing. Declining margins over time reveal unnoticed price reductions, while drops in margin after cost increases indicate delayed price updates. Sudden or unexplained changes in margin expose inconsistent pricing decisions. Variations in margins across similar transactions help identify lack of standardization. Overall, it provides clear visibility into pricing performance, enabling timely corrections and more consistent, data-driven pricing.

Formula Used:

$$\frac{\text{Selling Price} - \text{Cost Price}}{\text{Selling Price}} \times 100$$

Fig 4 : Profit Margin Formula

where , Selling Price(SP) = price at which product is sold

Cost Price(CP) = price at which product is purchased

$$\text{Profit} = \text{SP} - \text{CP}$$

Steps For Profit Margin Calculation:

Step1 . Extract Cost Price (CP) from raw sales Data

Step2 . Extract Selling Price (SP) from raw sales Data

Step3 . Calculate Profit for each transaction

$$\text{Profit} = \text{Selling Price} - \text{Cost Price}$$

Step4 . Compute Profit Margin (%)

$$\text{Profit Margin (\%)} = (\text{Selling Price} - \text{Cost Price} / \text{Selling Price}) * 100$$

Step5 . Apply calculation across all transactions/products

Step6 . Analyze margin patterns

Results and Findings

The raw sales data and cleaned dataset was analyzed using different analytical methods to uncover key patterns related to demand, inventory, and pricing. Based on these analyses, important findings were identified. To present these insights clearly, various graphs have been used, including box plots created using both Google Sheets and Excel, helping visualize trends, variability, and data distribution effectively.

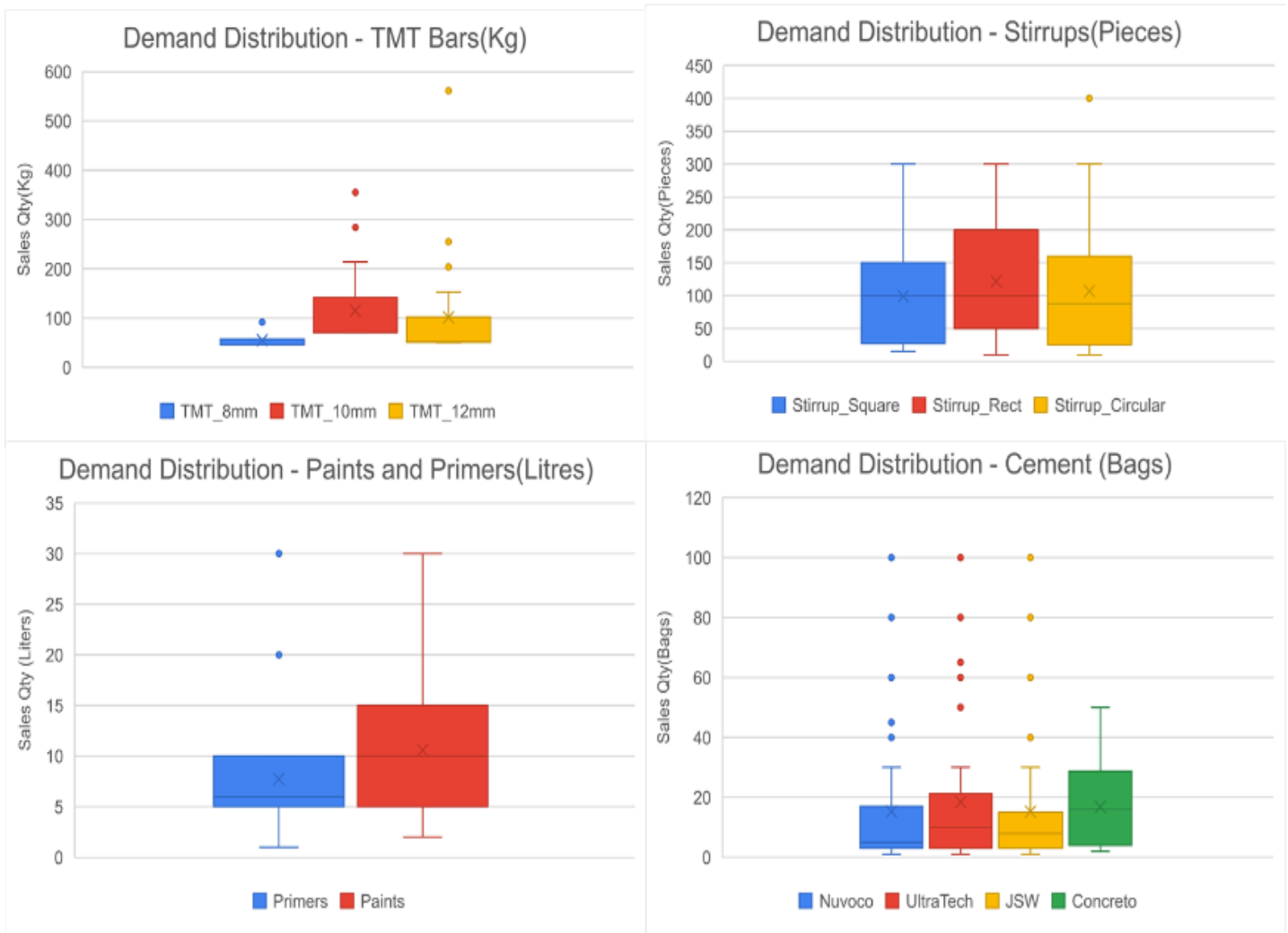


Fig 5 : Demand Distribution (Non-Zero Sales Days) Across Product Categories

Finding 1: Evidence of Demand Fluctuation Across Product Categories

The box plots below show demand distribution on actual sales days — meaning

zero-demand days are excluded.

- **Cements** - For all four cement brands, most of the sales are concentrated at the lower end, typically between 2 to 20 bags. However, there are a few high-value outliers where sales go up to 100 bags, indicating occasional bulk purchases.

For example, Nuvoco usually sells around 5 bags on a typical day, but there are instances where it goes up to 100 bags, showing a huge jump driven by these outliers. JSW and UltraTech follow a similar pattern — their typical sales are around 8 to 10 bags, but they also have extreme outliers reaching 100 bags.

Concreto behaves a bit differently. Its typical sales are slightly higher, around 16 bags, but it does not show such extreme outliers beyond 50 bags, which suggests that its demand is relatively more stable and predictable.

Another important point is that the variation in demand is quite high. The standard deviation for the main brands is around 15–16 bags, which is more than double the average demand of about 6 bags. This large variation is mainly due to the presence of these occasional high-value outliers.

- **TMT Bars** -TMT 8mm shows the most stable pattern among all types. In this analysis, quantities are recorded in kilograms (kg). The values are tightly clustered between 46 and 92 kg, which corresponds to approximately one to two bundles of **8mm rods (each bundle weighing around 42 kg and consisting of 10 rods)**. This indicates that most transactions involve small, consistent purchases. The limited spread is mainly because it was sold for only 10 days, so there isn't much variation in the data.

In contrast, TMT 10mm and 12mm behave very differently . TMT 12mm typically sells at around 51 kg, which corresponds to **one bundle of 12mm rods(consisting of 5 rods)**, indicating that most transactions involve a single bundle purchase. However, there are instances where sales reach up to 561 kg, which is approximately 11 bundles. This shows that while regular demand consists of small, consistent purchases, there are occasional bulk orders involving multiple bundles. These large orders create significant variation in demand.

This is also reflected in its high standard deviation (77.58 kg), which is almost double its average demand (42.4 kg), showing significant variability.

TMT 10mm follows a similar pattern. The typical sale is around **71 kg**, which corresponds to **one bundle of 10mm rods(consisting of 7 rods)**, indicating that most transactions involve a single bundle purchase. but it can go up to 355 kg in some cases. The high standard deviation (65.2 kg) again indicates that demand is not consistent and varies widely.

- **Stirrups** -All three stirrup types show a wide spread in their box plots, meaning that even on normal days, the number of pieces sold varies a lot. Rectangular stirrups have the highest typical demand (median of 100 pieces) and also the widest range, going from 10 to 450 pieces, showing very large fluctuations. Circular stirrups range from 10 to 400 pieces, and square stirrups from 15 to 300 pieces, both also showing significant variation.

The standard deviations (84, 75, and 60 pieces) are quite high, indicating that demand is not stable. The wide boxes (IQR) and lack of tight clustering suggest that stirrup demand is largely driven by specific construction needs rather than regular daily sales.

- **Paints and Primers (Litres)** These are the store's most stable products and the box plots reflect that clearly. The boxes are compact — Primer sits between roughly 3–10 litres with a median of 6 litres, and Paint between 5–20 litres with a median of 10 litres. Outliers reach 30 litres but they are rare. Standard deviations of 5.1 (Primer) and 6.4 (Paint) are actually close to their means of 3.1 and 3.4 litres, making these the only products where average demand is a reasonably reliable planning number.

Finding 2: Evidence of Inventory Fluctuation Using Product-wise ROP Analysis

- **Cement**

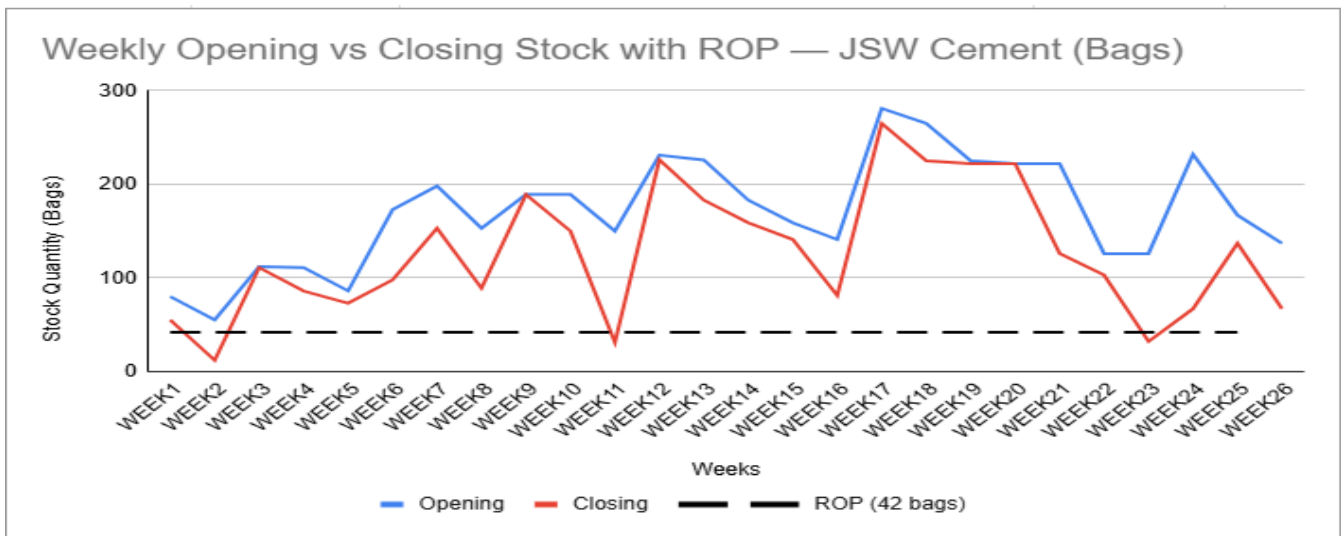


Fig 6: Weekly Opening vs Closing Stock with Reorder Point (ROP) — JSW Cement (Bags)

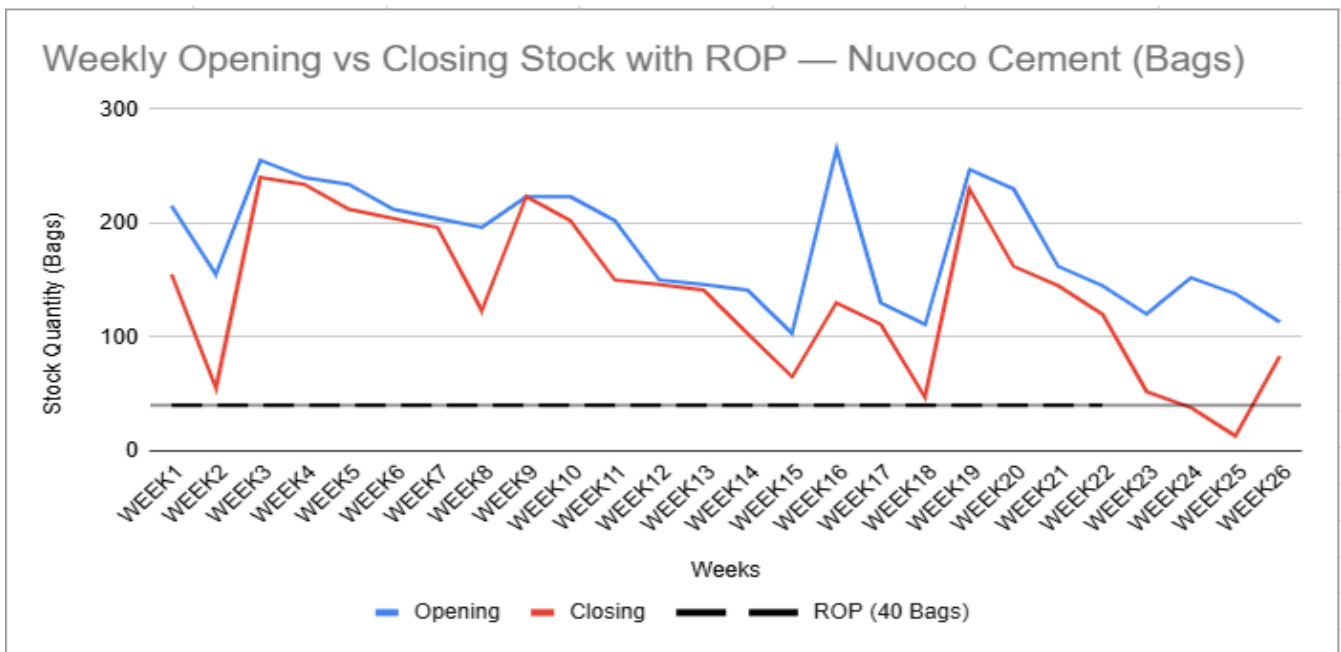


Fig 7: Weekly Opening vs Closing Stock with Reorder Point (ROP) — Nuvoco Cement (Bags)

JSW Cement: Closing stock falls below the ROP level of 42 bags in **3 out of 26 weeks**, indicating occasional stockout risk. While most weeks remain above the safety threshold, these few breaches highlight periods where inventory was insufficient to meet demand.

Nuvoco Cement: It has a ROP equal to 40 and it performs slightly better, with only **2 out of 26 weeks** falling below the ROP. Most of the time, both opening and closing stock remain well

above the safety threshold , indicating more consistent inventory control with fewer critical shortages.

UltraTech and Concreto Cement: UltraTech Cement (ROP = 40 bags) shows only 1 out of 26 weeks (3.8%) below the ROP, while Concreto Cement (ROP = 34 bags) shows 0 stockout instances (0/26 weeks), indicating highly stable and well-managed inventory levels.

These products are not displayed in the above figure but are included in the **Visuals table for reference**, where their low stockout frequency further supports their reliability compared to other cement brands.

- **TMT Bars**

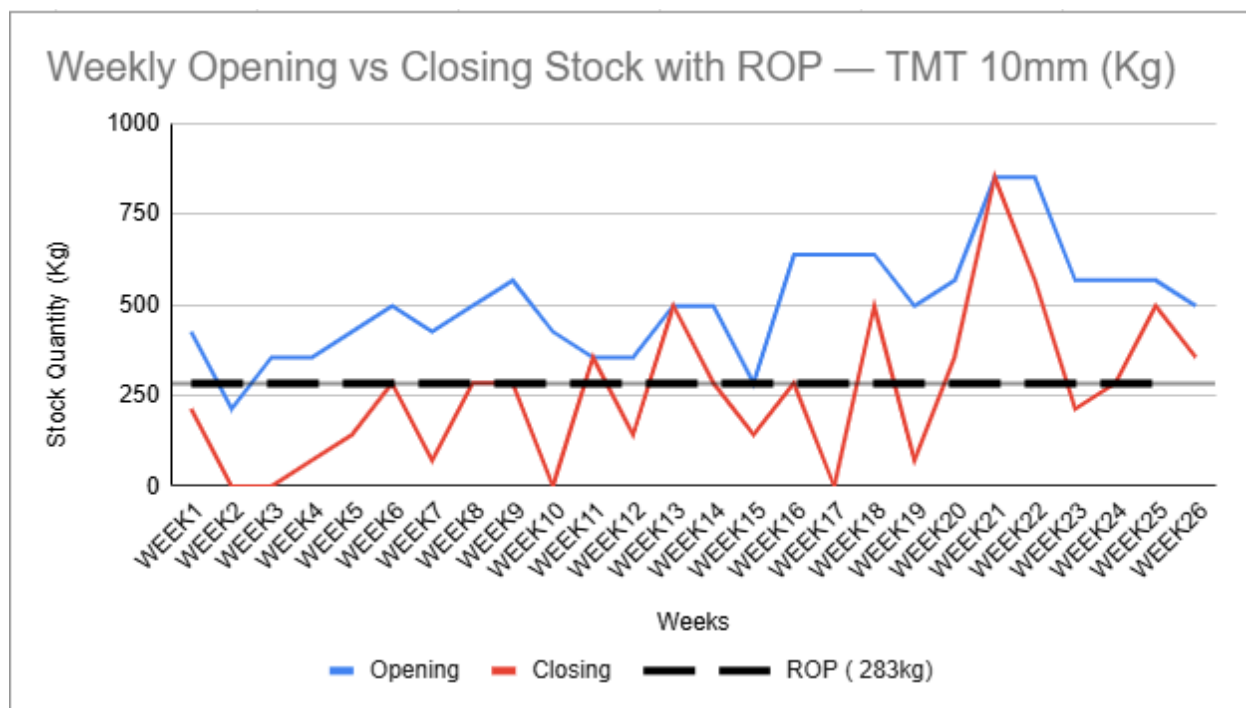


Fig 8 : Weekly Opening vs Closing Stock with Reorder Point (ROP) — TMT 10mm (Kg)

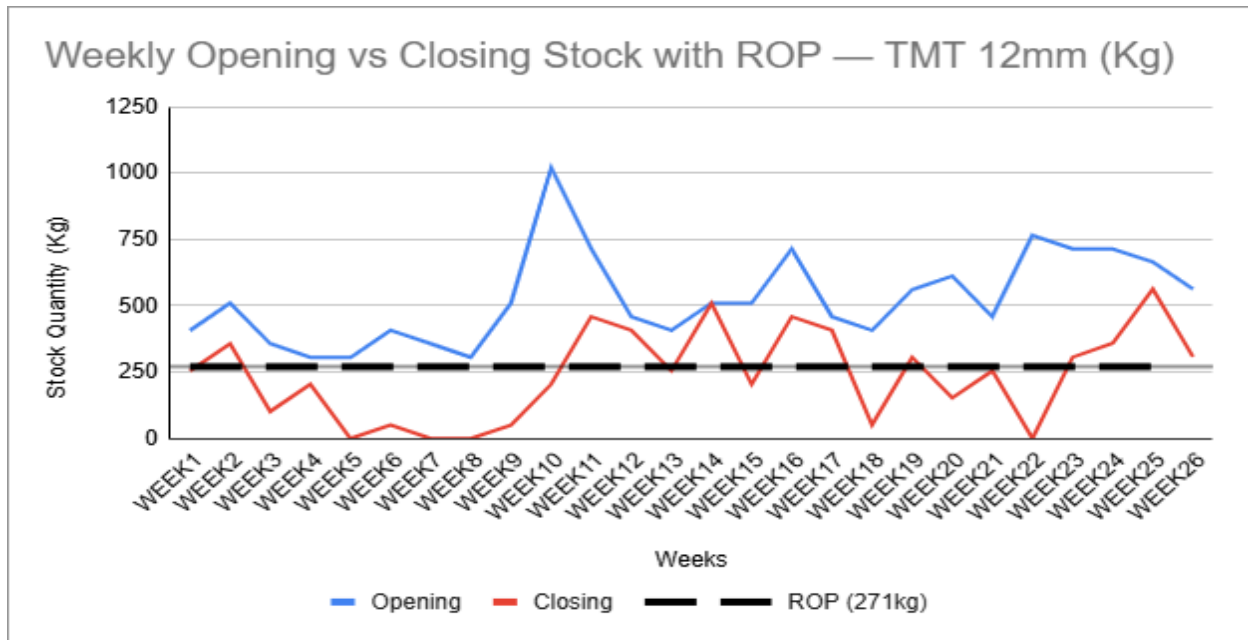


Fig 9: Weekly Opening vs Closing Stock with Reorder Point (ROP) — TMT 12mm (Kg)

TMT 10mm (ROP = 283 kg): Closing stock falls below ROP in **12/26 weeks (~46%)**, clearly identifying frequent stockout situations.

TMT 12mm (ROP ≈ 271 kg): Closing stock remains below ROP in ~15/26 weeks (~58%), highest risk among TMT types.

TMT 8mm (ROP = 82 kg): Stock remains mostly above ROP (only 2/26 weeks below ~7.7%), showing stable demand.
(available in visuals and calculations table for reference)

- **Stirrups**

Stirrup Rectangular (ROP ≈ 398 pcs):

Closing stock falls below ROP in 11/26 weeks (42.3%).

ROP identifies frequent reorder delays, indicating need for earlier replenishment.

Stirrup Circular (ROP $\approx$ 294 pcs):

Falls below ROP in 9/26 weeks (34.6%).

ROP highlights multiple risk weeks, enabling timely stock planning.

Stirrup Square (ROP $\approx$ 200 pcs):

Falls below ROP in 7/26 weeks (26.9%).

ROP confirms moderate risk, requiring controlled monitoring.

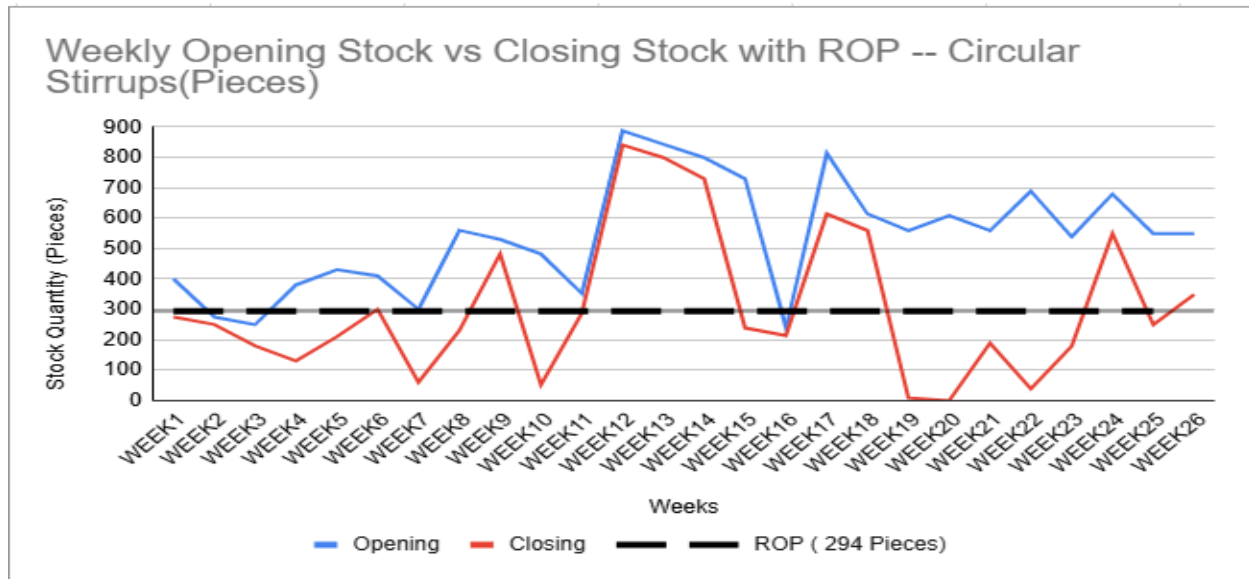


Fig 10: Weekly Opening vs Closing Stock with Reorder Point (ROP) — Circular Stirrups(Pieces)

(refer to calculations and visuals table for square stirrups and rectangle stirrups)

- **Finding 3 : Pricing Ambiguity**

3.1 : Profit margin analysis identified price ambiguity in Stirrup_Circular, where the same product was sold at multiple prices (₹8, ₹9, ₹10) without a consistent pricing rule. In 9 transactions (Oct 2025–Jan 2026), 474 pieces were sold at cost price (0% margin). These transactions occurred across different months, involved different customers(verified through bills) , and were not bulk purchases, confirming that the variation is not demand-driven. In contrast, Stirrup_Rect and Stirrup_Square consistently

maintain margins above 30%, further highlighting the inconsistency in pricing strategy within the same product category.

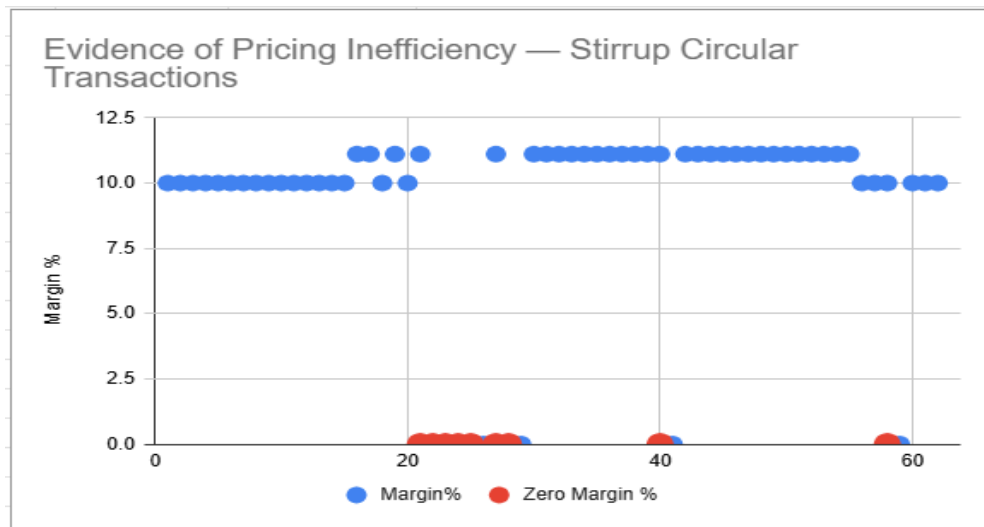


Fig 11: Pricing Inefficiency of circular stirrup

3.2 : The scatter plot reveals clear price ambiguity in UltraTech cement sales, where multiple selling prices were recorded within the same month despite similar cost conditions. In October, the product was sold at both ₹330 and ₹340 per bag, while in November, three different prices — ₹320, ₹330, and ₹340 — were charged within the same month.

These transactions were made to different customers and are verified through individual sales bills, confirming that the variation is not a data error.

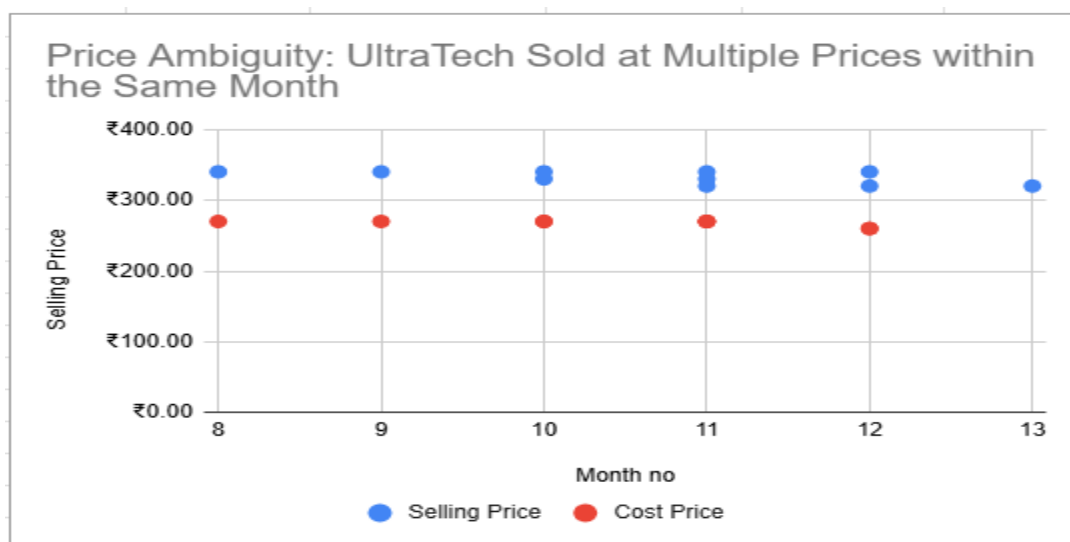


Fig 12: UltraTech Cement Sold at Multiple Prices within the Same Month

Interpretation of Results and Recommendations

Problem1: Demand Fluctuation

Interpretation:

- **Cement:** Demand is spike-driven — mostly 2–10 bags, but occasional bulk orders make the average unreliable for planning.
- **TMT Bars:** Demand is highly variable, especially for 10mm and 12mm — most transactions are moderate, but some reach very high quantities, creating sharp fluctuations.
- **Stirrups:** Highly volatile — quantities vary widely even on normal days, making fixed stocking ineffective.
- **Paints & Primers:** Stable demand — low variation makes average-based inventory planning reliable.

Recommendations:

1. Category-Specific Safety Stock Planning: Use buffer stock for high-variability products (cement, TMT, stirrups) instead of average; keep paints and primers on fixed reorder levels due to stable demand.

2. Bulk Demand Tracking: Maintain a log of large transactions (cement >50 bags, TMT >200 kg, stirrups >200 pcs) to identify patterns and plan inventory in advance.

Problem2: Inventory Management

Interpretation:

- **TMT 12mm:** 12mm TMT bar was below ROP in 15/26 weeks, with stock reaching zero/negative (min –255 kg), indicating frequent stockouts and unmet demand.
- **TMT 10mm:** Similar pattern — below ROP in 12/26 weeks with multiple stockouts (min –142 kg), showing inconsistent replenishment cycles.
- **Cement Brands:** Relatively stable — minimal stockouts (0–3 weeks), indicating better inventory control and serving as a benchmark.

Recommendations:

- 1. Implement weekly ROP checks every Weekend.**
- 2. Prioritise TMT Bars and Stirrups for immediate reorder discipline**

Problem3: Price Ambiguity

Interpretation:

- Stirrup Circular was sold at 0% margin in 9 transactions — involving 474 pieces across Oct 2025 to Jan 2026. These occurred in different months, to different customers (verified from bills), and were not bulk discounts. The selling price was set exactly equal to cost price (₹8 = ₹8), generating ₹0 profit on ₹3,792 in revenue.
- UltraTech was sold at three different prices in a single month — ₹320, ₹330, and ₹340 were all charged in November 2025 despite a stable cost price of ₹270. The scatter plot confirms this is not a data error — different customers, same product, same cost, different price.

Recommendations:

- **Set a minimum profit margin on every sale:**
Ensure that no product is sold at or near cost price by maintaining a minimum margin on all transactions.
- **Update prices when costs change:**
Whenever the cost price changes, revise the selling price within 2–3 days and keep a record of why the change was made.
- **Review margins regularly:**
At the end of each month, check the profit margins for all products to identify any issues early and take corrective action.

Additional

Dataset Link

 FinalData_22f1001112 (1).xlsx